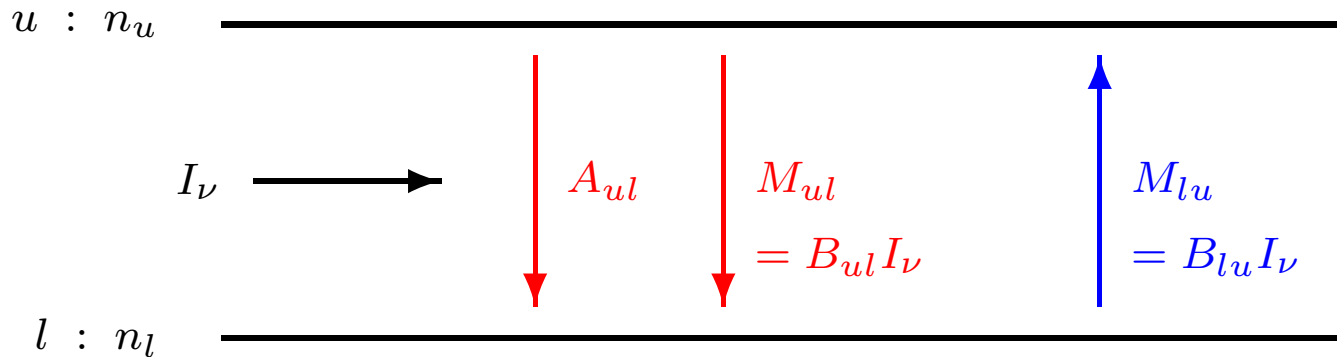

Lecture 11: Line Emission Mechanisms

Chin-Fei Lee

`cflee@asiaa.sinica.edu.tw`

Line Emission Mechanisms

Consider a two-level system with only radiative processes:



$$\text{(Spontaneous Emission)} \quad dE_e = \frac{h\nu_0}{4\pi} n_u A_{ul} \phi_e(\nu) dV d\Omega d\nu dt \quad (\text{ergs})$$

$$\text{(Absorption)} \quad dE_a = \frac{h\nu_0}{4\pi} n_l M_{lu} \phi_e(\nu) dV d\Omega d\nu dt \quad (\text{ergs})$$

$$\text{(Stimulated Emission)} \quad dE_s = \frac{h\nu_0}{4\pi} n_u M_{ul} \phi_e(\nu) dV d\Omega d\nu dt \quad (\text{ergs})$$

Here A_{ul} , M_{ul} , and M_{lu} are **transition probabilities** and all have a unit of s^{-1} , dV is infinitesimal volume. Thus, with the infinitesimal area $d\sigma$, the change in energy along the path is given by

$$dE = dI_\nu d\Omega d\sigma d\nu dt = dE_e + dE_s - dE_a \quad (\text{ergs})$$

Radiative Transfer of Line Emission

From previous slide:

$$dI_\nu d\Omega d\sigma d\nu dt = dE_e + dE_s - dE_a \quad (\text{ergs})$$

Assuming the line profiles $\phi_e(\nu) = \phi_s(\nu) = \phi_a(\nu) \equiv \phi(\nu)$, then

$$dI_\nu d\Omega d\sigma d\nu dt = \frac{h\nu_0}{4\pi} [n_u A_{ul} + n_u M_{ul} - n_l M_{lu}] \phi(\nu) d\Omega dV d\nu dt$$

With $dV = d\sigma ds$, i.e., $d\sigma = dV/ds$, then

$$\begin{aligned} \frac{dI_\nu}{ds} &= \frac{h\nu_0}{4\pi} [n_u A_{ul} + n_u M_{ul} - n_l M_{lu}] \phi(\nu) \\ &= \frac{h\nu_0}{4\pi} [n_u A_{ul} + (n_u B_{ul} - n_l B_{lu}) I_\nu] \phi(\nu) \\ &\equiv -\alpha_\nu I_\nu + j_\nu \end{aligned}$$

with

$$\alpha_\nu = \frac{h\nu_0}{4\pi} [n_l B_{lu} - n_u B_{ul}] \phi(\nu) \quad \text{and} \quad j_\nu = \frac{h\nu_0}{4\pi} n_u A_{ul} \phi(\nu)$$

Relations betw. Einstein Coefficients

Let us define a mean intensity

$$J_\nu = \frac{1}{4\pi} \int I_\nu d\Omega$$

and

$$\bar{J} = \int J_\nu \phi(\nu) d\nu \quad \text{with} \quad \int \phi(\nu) d\nu \equiv 1$$

If [the material is in thermodynamical equilibrium with the radiation](#), then the number of transitions per unit time per unit volume out of state u = the number of transitions per unit time per unit volume into state u , i.e.,

$$n_u A_{ul} + n_u B_{ul} \bar{J} = n_l B_{lu} \bar{J} \quad \Rightarrow \quad \bar{J} = \frac{A_{ul}/B_{ul}}{(n_l/n_u)(B_{lu}/B_{ul}) - 1}$$

In addition, if [the material is in thermodynamical equilibrium with itself](#), we also have

$$\frac{n_u}{n_l} = \frac{g_u}{g_l} e^{-h\nu_0/kT}$$

Therefore, we have

$$\bar{J} = \frac{A_{ul}/B_{ul}}{(g_l/g_u)(B_{lu}/B_{ul}) \exp(h\nu_0/kT) - 1}$$

Relations betw. Einstein Coefficients

From previous slide,

$$\bar{J} = \frac{A_{ul}/B_{ul}}{(g_l/g_u)(B_{lu}/B_{ul}) \exp(h\nu_0/kT) - 1}$$

In thermodynamical equilibrium, we also have $I_\nu = B_\nu$ and thus $\bar{J} = B_\nu$ and

$$\bar{J} = B_\nu = \frac{2h\nu^3}{c^2} \frac{1}{\exp(h\nu_0/kT) - 1}$$

Comparing the above two eqs., we have

$$A_{ul} = \frac{2h\nu_0^3}{c^2} B_{ul}$$

and

$$g_l B_{lu} = g_u B_{ul}$$

Thus, for given A_{ul} (s^{-1}), we can derive B_{ul} and then B_{lu} ($\text{sr cm}^2 \text{ Hz erg}^{-1}$).

Absorption Coefficient

Since we are usually given with A_{ul} , it is better to express the absorption coefficient in term of A_{ul} .
From previous slide,

$$\alpha_\nu = \frac{h\nu_0}{4\pi} [n_l B_{lu} - n_u B_{ul}] \phi(\nu) = \frac{h\nu_0}{4\pi} n_l B_{lu} \left[1 - \frac{n_u B_{ul}}{n_l B_{lu}} \right] \phi(\nu)$$

Since

$$B_{lu} = \frac{g_u}{g_l} B_{ul} = \frac{g_u}{g_l} \frac{c^2}{2h\nu_0^3} A_{ul}$$

we have

$$\alpha_\nu = \frac{c^2}{8\pi\nu_0^2} \frac{g_u}{g_l} n_l A_{ul} \left[1 - \frac{n_u g_l}{n_l g_u} \right] \phi(\nu)$$

In LTE, we have

$$\frac{n_u}{n_l} = \frac{g_u}{g_l} \exp\left(-\frac{h\nu_0}{kT}\right)$$

and thus

$$\alpha_\nu = \frac{c^2}{8\pi\nu_0^2} \frac{g_u}{g_l} n_l A_{ul} \left[1 - \exp\left(-\frac{h\nu_0}{kT}\right) \right] \phi(\nu)$$

Emission for a simple electric dipole

Let us study the [Hertz dipole](#). It is a simple electric dipole with the charge motion given by

$$x(t) = x_0 \exp(-i\omega t)$$

The total amount of power radiated by the dipole is (see previous lecture)

$$P(t) = \int |S| r^2 d\Omega = \frac{2}{3} \frac{e^2 \dot{v}^2(t)}{c^3}$$

Since

$$\dot{v} = \ddot{x} = -\omega^2 x_0 \exp(-i\omega t) = -4\pi^2 \nu^2 x_0 \exp(-i\omega t)$$

we have

$$P(t) = \frac{32\pi^4 \nu^4}{3} \frac{e^2 x_0^2}{c^3} \exp^2(-i\omega t)$$

The time-average of the exponential term is $\frac{1}{2}$, thus the [mean power](#) is

$$\langle P \rangle = \frac{64\pi^4 \nu^4}{3c^3} \left(\frac{ex_0}{2} \right)^2$$

Emission for a simple electric dipole

From previous slide

$$\langle P \rangle = \frac{64\pi^4\nu^4}{3c^3} \left(\frac{ex_0}{2} \right)^2$$

If we write this in terms of A_{ul} , then

$$\langle P \rangle = h\nu_{ul}A_{ul}$$

Therefore,

$$A_{ul} = \frac{64\pi^4}{3hc^3} \nu_{ul}^3 \left(\frac{ex_0}{2} \right)^2 = \frac{64\pi^4}{3hc^3} \nu_{ul}^3 |\mu_{ul}|^2$$

where

$$\mu_{ul} = \frac{ex_0}{2}$$

is the mean electric dipole moment. For the transition probability for atomic hydrogen close to the Lyman limit, which has $\nu = 3.29 \times 10^{15}$ Hz, is 10^9 s^{-1} .

Emission for a magnetic dipole

Two sources of magnetic moment: (1) (orbital) motion of electric charges, such as electric currents and (2) the intrinsic magnetism of elementary particles, such as the electron (spin). (Both associated with Angular momentum). For a magnetic dipole,

$$m(t) = m_0 \exp(-i\omega t)$$

the corresponding Einstein A coefficient is

$$A_{ul} = \frac{64\pi^4}{3hc^3} \nu_{ul}^3 |\mu_{ul}^*|^2$$

where μ_{ul}^* is the mean magnetic dipole moment of the oscillator for this transition. For a hydrogen atom with the electron at the lowest Bohr orbit, the magnetic dipole moment is mainly from the spin of the electron:

$$|\mu_{ul}^*| = \frac{e\hbar}{2m_e c} \cong 9.27 \times 10^{-21} \text{ erg Gauss}^{-1}$$

Thus, for the freq at the Lyman limit $\nu = 3.29 \times 10^{15} \text{ Hz}$,

$$A_{ul} \cong 10^4 \text{ s}^{-1}$$

This transition probability is smaller than that of an electric dipole by a factor of 2092.

Line Emission: Rate Equation

In general, we need to consider (1) multi-level, (2) not only radiative, but also collisional, etc processes, and (3) non-LTE. What we have are

$$\frac{dn_l}{dt} = -n_l \sum_m \sum_i R_{lm}^i + \sum_m n_m \sum_i R_{ml}^i$$

and

$$\sum_l n_l = n_{\text{tot}}$$

with

R_{ml}^i : transition rate between level m and l through process i

For equilibrium,

$$\frac{dn_l}{dt} = 0$$

Simple Examples: 2-level system

First we consider the case of two states, l and u , where the only way to change states is by the emission and absorption of radiation. We do not however, assume that LTE applies. The transition rates are given by the Einstein coefficients. For a stationary situation, we have

$$n_u A_{ul} + n_u B_{ul} \bar{J} = n_l B_{lu} \bar{J} \quad \Rightarrow \quad \frac{n_u}{n_l} = \frac{B_{lu}}{B_{ul}} \frac{\bar{J}}{\frac{2h\nu_0^3}{c^2} + \bar{J}}$$

We can define a **brightness temperature** T_b for the radiation:

$$\bar{J} = \frac{2h\nu_0^3}{c^2} \frac{1}{\exp(h\nu_0/kT_b) - 1}$$

which can be rewritten as

$$\frac{\bar{J}}{\bar{J} + \frac{2h\nu_0^3}{c^2}} = \exp\left(-\frac{h\nu_0}{kT_b}\right) \quad \text{or} \quad T_b = \frac{h\nu_0}{k} \frac{1}{\ln\left(\frac{2h\nu_0^3}{c^2\bar{J}} + 1\right)}$$

Thus,

$$\frac{n_u}{n_l} = \frac{g_u}{g_l} \exp\left(-\frac{h\nu_0}{kT_b}\right)$$

Simple Examples

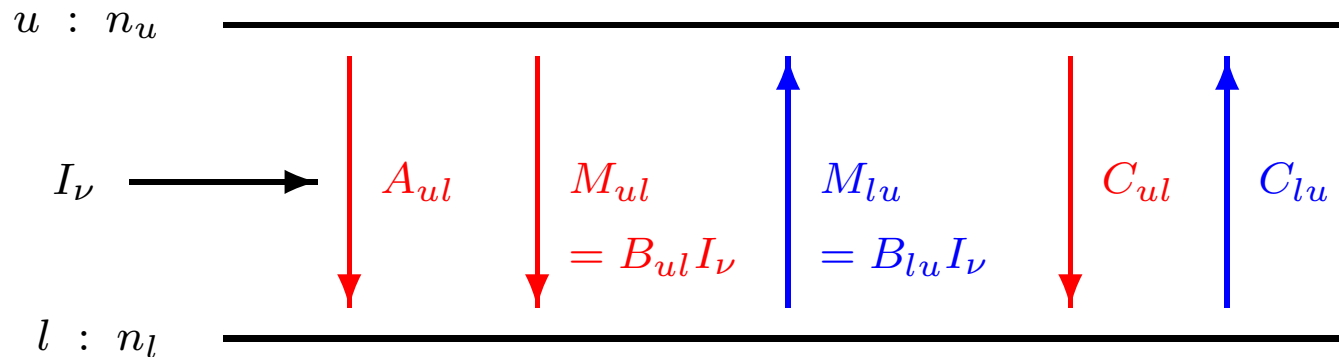
From previous slide,

$$\frac{n_u}{n_l} = \frac{g_u}{g_l} \exp\left(-\frac{h\nu_0}{kT_b}\right)$$

The number densities n_u and n_l thus will be described by a Boltzmann distribution as in the case of LTE. The temperature T_b in this distribution describes the **radiation density** ($= \frac{4\pi}{c} \bar{J}$) at the frequency corresponding to the transition $u \rightarrow l$. This need have nothing to do with a thermodynamic temperature of the system.

Simple Examples

Another simple example for finding the number densities of the cool parts of the ISM gas. In this case, collision probabilities are needed for the transition rates.



Here C_{ul} and C_{lu} are the collision probabilities per particle (in $\text{cm}^3 \text{s}^{-1}$) increasing with increasing density and with

$$\frac{C_{lu}}{C_{ul}} = \frac{g_u}{g_l} \exp\left(-\frac{h\nu_0}{kT_K}\right)$$

where T_K is the kinetic temperature. In equilibrium, we have

$$n_u(A_{ul} + B_{ul}\bar{J} + C_{ul}) = n_l(B_{lu}\bar{J} + C_{lu}) \Rightarrow \frac{n_u}{n_l} = \frac{B_{lu}\bar{J} + C_{lu}}{A_{ul} + B_{ul}\bar{J} + C_{ul}}$$

Various Temperatures

Since

$$\bar{J} = \frac{2h\nu_0^3}{c^2} \frac{1}{\exp(h\nu_0/kT_b) - 1}$$

we have

$$B_{lu}\bar{J} = \frac{g_u}{g_l} \frac{c^2}{2h\nu^3} A_{ul} = \frac{g_u}{g_l} \frac{A_{ul}}{\exp(h\nu_0/kT_b) - 1} = \frac{g_u}{g_l} \frac{A_{ul} \exp(-h\nu_0/kT_b)}{1 - \exp(-h\nu_0/kT_b)}$$

and

$$A_{ul} + B_{ul}\bar{J} = A_{ul} + \frac{g_l}{g_u} B_{lu}\bar{J} = \frac{A_{ul}}{1 - \exp(-h\nu_0/kT_b)}$$

Also

$$C_{lu} = C_{ul} \frac{g_u}{g_l} \exp\left(-\frac{h\nu_0}{kT_K}\right)$$

Therefore,

$$\frac{n_u}{n_l} = \frac{B_{lu}\bar{J} + C_{lu}}{A_{ul} + B_{ul}\bar{J} + C_{ul}} = \frac{g_u}{g_l} \exp\left(-\frac{h\nu_0}{kT_b}\right) \frac{A_{ul} + C_{ul} \exp\left(-\frac{h\nu_0}{kT_K}\right) [\exp\left(\frac{h\nu_0}{kT_b}\right) - 1]}{A_{ul} + C_{ul} [1 - \exp\left(-\frac{h\nu_0}{kT_b}\right)]}$$

Various Temperatures

Let us define an **excitation temperature** T_{ex} for which

$$\frac{n_u}{n_l} \equiv \frac{g_u}{g_l} \exp\left(-\frac{h\nu_0}{kT_{\text{ex}}}\right)$$

then

$$\exp\left(-\frac{h\nu_0}{kT_{\text{ex}}}\right) = \exp\left(-\frac{h\nu_0}{kT_b}\right) \frac{A_{ul} - C_{ul} \exp\left(-\frac{h\nu_0}{kT_K}\right) [1 - \exp\left(-\frac{h\nu_0}{kT_b}\right)]}{A_{ul} + C_{ul} [1 - \exp\left(-\frac{h\nu_0}{kT_b}\right)]}$$

This excitation temperature is a mean between the radiation temperature T_b and the kinetic temperature T_K , i.e., $T_b \leq T_{\text{ex}} \leq T_K$. Let us define

$$T_0 = \frac{h\nu_0}{k}$$

then when T_{ex} , T_b and $T_K \gg T_0$, we have

$$T_{\text{ex}} \approx T_K \frac{T_b A_{ul} + T_0 C_{ul}}{T_K A_{ul} + T_0 C_{ul}}$$

Various Temperatures

From previous slide

$$T_{\text{ex}} \approx T_K \frac{T_b A_{ul} + T_0 C_{ul}}{T_K A_{ul} + T_0 C_{ul}}$$

(1) if radiation dominates the rate eq. ie., $A_{ul} \gg C_{ul}$, then $T_{\text{ex}} \rightarrow T_b$.

(2) if collision dominates the rate eq. ie., $C_{ul} \gg A_{ul}$, then $T_{\text{ex}} \rightarrow T_K$.

Since C_{ik} increases with increasing density N , collisions will dominate the distribution in high-density situation and the excitation temperature of the line will be equal to the kinetic temperature. In low density situation $T_{\text{ex}} \rightarrow T_b$. The density when $A_{ul} \approx C_{ul} \approx N^* \langle \sigma v \rangle$ is called critical density. The smaller A_{ul} , the lower is N^* .

In general, Non-LTE and

$$T_b < T_{\text{ex}} < T_K$$

In the case of LTE, we have

$$T_b = T_{\text{ex}} = T_K = T$$

and

$$I = B(T_b) = B(T), \quad \frac{n_u}{n_l} = \frac{g_u}{g_l} \exp\left(-\frac{h\nu_0}{kT_{\text{ex}}}\right) = \frac{g_u}{g_l} \exp\left(-\frac{h\nu_0}{kT}\right)$$

Absorption and Emission Coeff.

Once **excitation temperature** T_{ex} is derived/given, then since (from previous slide)

$$j_{\nu} = \frac{h\nu_0}{4\pi} n_u A_{ul} \phi(\nu)$$

$$\alpha_{\nu} = \frac{c^2}{8\pi\nu_0^2} \frac{g_u}{g_l} n_l A_{ul} \left[1 - \frac{n_u g_l}{n_l g_u} \right] \phi(\nu)$$

we have

$$j_{\nu} = \frac{h\nu_0}{4\pi} \frac{n_l g_u}{g_l} \exp\left(-\frac{h\nu_0}{kT_{\text{ex}}}\right) A_{ul} \phi(\nu)$$

and

$$\alpha_{\nu} = \frac{c^2}{8\pi\nu_0^2} \frac{g_u}{g_l} n_l A_{ul} \left[1 - \exp\left(-\frac{h\nu_0}{kT_{\text{ex}}}\right) \right] \phi(\nu)$$

Remember them because they can be used to derive other quantities, like column density and opacity.

Emissivity

Generally, since

$$B_{ul} = \frac{A_{ul}}{S_0} \quad \text{with } S_0 \equiv \frac{2h\nu_0^3}{c^2}$$

and

$$B_{lu} = \frac{g_u}{g_l} B_{ul} = \frac{g_u}{g_l} \frac{A_{ul}}{S_0}$$

Then we also have

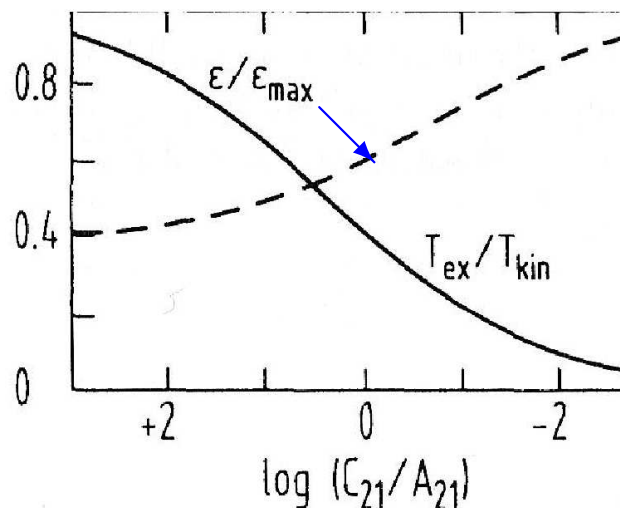
$$\frac{n_u}{n_l} = \frac{B_{lu}\bar{J} + C_{lu}}{A_{ul} + B_{ul}\bar{J} + C_{ul}} = \frac{g_u}{g_l} \frac{\bar{J}/S_0 + C_{ul}/A_{ul} \exp(-T_0/T_K)}{1 + \bar{J}/S_0 + C_{ul}/A_{ul}}$$

As expected, as $C_{ul}/A_{ul} \gg 1$, the population of the two-level system is determined by the kinematic temperature. The emissivity can then also be written as

$$\begin{aligned} j_\nu &= \frac{h\nu_0}{4\pi} n_u A_{ul} \phi(\nu) \\ &= \frac{h\nu_0}{4\pi} \frac{g_u n_l}{g_l} \frac{\bar{J}/S_0 + C_{ul}/A_{ul} \exp(-T_0/T_K)}{1 + \bar{J}/S_0 + C_{ul}/A_{ul}} A_{ul} \phi(\nu) \end{aligned}$$

Emissivity

$$j_\nu = \frac{h\nu_0}{4\pi} \frac{g_u n_l}{g_l} \frac{\bar{J}/S_0 + C_{ul}/A_{ul} \exp(-T_0/T_K)}{1 + \bar{J}/S_0 + C_{ul}/A_{ul}} A_{ul} \phi(\nu)$$



In the limit of small C_{ul}/A_{ul} (i.e., low N), that is, **a very sub thermally excited state** ($T_{\text{ex}} \ll T_K$ or **very low n_u/n_l**), there will still be some emission. For a detectable line however, the abundance must be sufficiently large. E.g., CO has a small A coefficient ($6.5 \times 10^{-8} \text{ s}^{-1}$ for $J=1-0$) as compared to others (e.g., 3.8×10^{-5} for CS $J=2-1$ and $2.2 \times 10^{-5} \text{ s}^{-1}$ for C^{18}O $J=1-0$, see [Table 14.2](#)) and thus low critical density, therefore the levels are populated at densities of a few 100 cm^{-3} . But it is abundant and thus has high emissivity. The total emission also depends on column density. Emission from the $J=1-0$ and $2-1$ lines of CO is widespread also because of the large column density.

Atomic line: HI 21 cm line

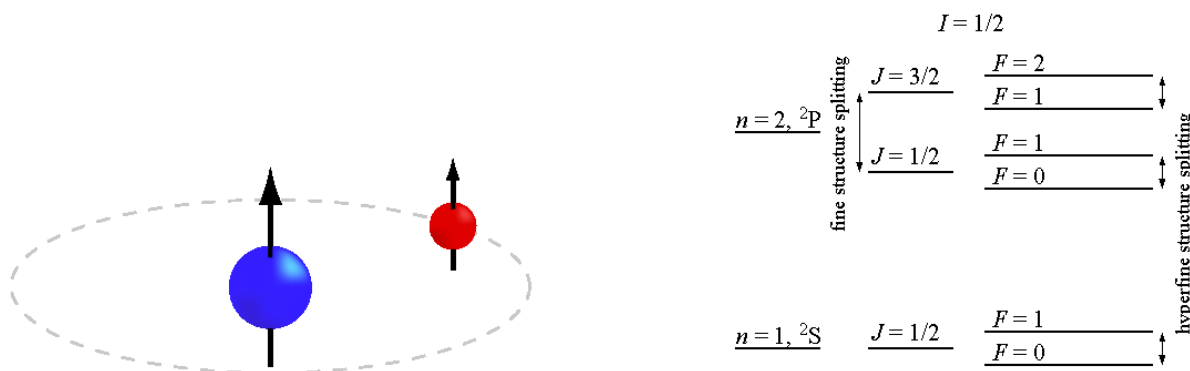
- first radio wavelength spectral line detected
- prediction: (1945) Dutch astronomers Van de Hulst and his teacher Oort
- first detection: Ewen and Purcell (1951)



This horn antenna, now displayed in front of the Jansky Lab at NRAO in Green Bank, WV, was used by Harold Ewen and Edward Purcell, then at the Lyman Laboratory of Harvard University, in the first detection of the 21 cm emission from [neutral hydrogen in the Milky Way](#). The emission was first detected on March 25, 1951.

Atomic line: HI 21 cm line

- Hyperfine splitting - interaction between the magnetic moment (spin) of the proton and the magnetic moment (spin) of the orbiting electron at the ground state



(Left) An electron orbiting a proton with parallel spins ($F = 1$) has higher energy than if the spins were anti-parallel ($F = 0$). One 21 cm photon is emitted when the spins flip from parallel to antiparallel. (Right) Fine and hyperfine structure in hydrogen. The hyperfine splitting of the ground $n = 1, 2S$ state is the source of the 21 cm hydrogen line.

- Rest Frequency: $1.420405751786 \pm 0.000000000030$ MHz
- Einstein A ($2.86888 \times 10^{-15} \text{ s}^{-1}$) is **very small** (i.e., $t_{1/2} = 1/A \sim 10^7$ yr) due to
 - small (magnetic) dipole transition (Note electron spin $L = \hbar/2$)

$$|\mu_{ul}^*| = \frac{\hbar}{2} \frac{e}{m_e c} \cong 9.27 \times 10^{-21} \text{ erg Gauss}^{-1}$$

- low frequency

Excitation Temp. of HI 21 cm line

For the 21 cm line, the excitation temperature T_{ex} generally called “spin” temperature T_s , thus

$$\frac{n_u}{n_l} = \frac{g_u}{g_l} \exp\left(-\frac{h\nu_0}{kT_s}\right) = 3 \exp\left(-\frac{T_0}{T_s}\right)$$

where $g_u = 3$, $g_l = 1$, ν_0 is the rest frequency of the 21cm line and

$$T_0 = \frac{h\nu_0}{k} = 0.0682 \text{ K}$$

Recall that

$$T_s \equiv T_{\text{ex}} \approx T_K \frac{T_b A_{ul} + T_0 C_{ul}}{T_K A_{ul} + T_0 C_{ul}}$$

Defining

$$y = \frac{h\nu}{kT_K} \frac{C_{ul}}{A_{ul}}$$

then

$$T_s \approx \frac{T_b + yT_K}{1 + y} \approx T_K \quad \because y \gg 1$$

Absorption Coeff. of HI 21 cm line

Recall that the absorption coefficient can be written as

$$\alpha_\nu = \frac{c^2}{8\pi\nu_0^2} \frac{g_u}{g_l} n_l A_{ul} \left[1 - \exp\left(-\frac{T_0}{T_s}\right)\right] \phi(\nu)$$

From previous slide,

$$\frac{n_u}{n_l} = 3 \exp\left(-\frac{T_0}{T_s}\right)$$

In the ISM with $T_K \sim 10 - 10^4$ K, since $T_s \approx T_K \gg T_0$, we have

$$\frac{n_u}{n_l} \approx 3 \quad \Rightarrow \quad n_H = n_u + n_l \approx 4n_l$$

and

$$1 - \exp\left(-\frac{T_0}{T_s}\right) \approx \frac{T_0}{T_s} = \frac{h\nu_0}{kT_s}$$

then

$$\alpha_\nu \approx \frac{3c^2}{32\pi} \frac{n_H A_{ul}}{\nu_0} \frac{h}{kT_s} \phi(\nu)$$

Column density from HI 21 cm line

The neutral hydrogen **column density** along any line-of-sight is defined as

$$N_{\text{H}} \equiv \int_{\text{los}} n_{\text{H}}(s) ds \quad (\text{cm}^{-2})$$

Assuming **ISOTHERMAL** with the same T_s along the line of sight, then the line opacity is

$$\tau_{\nu} = \int \alpha_{\nu} ds \approx \frac{3c^2}{32\pi} \frac{A_{ul}}{\nu_0} \frac{h\phi(\nu)}{kT_s} N_{\text{H}}$$

thus

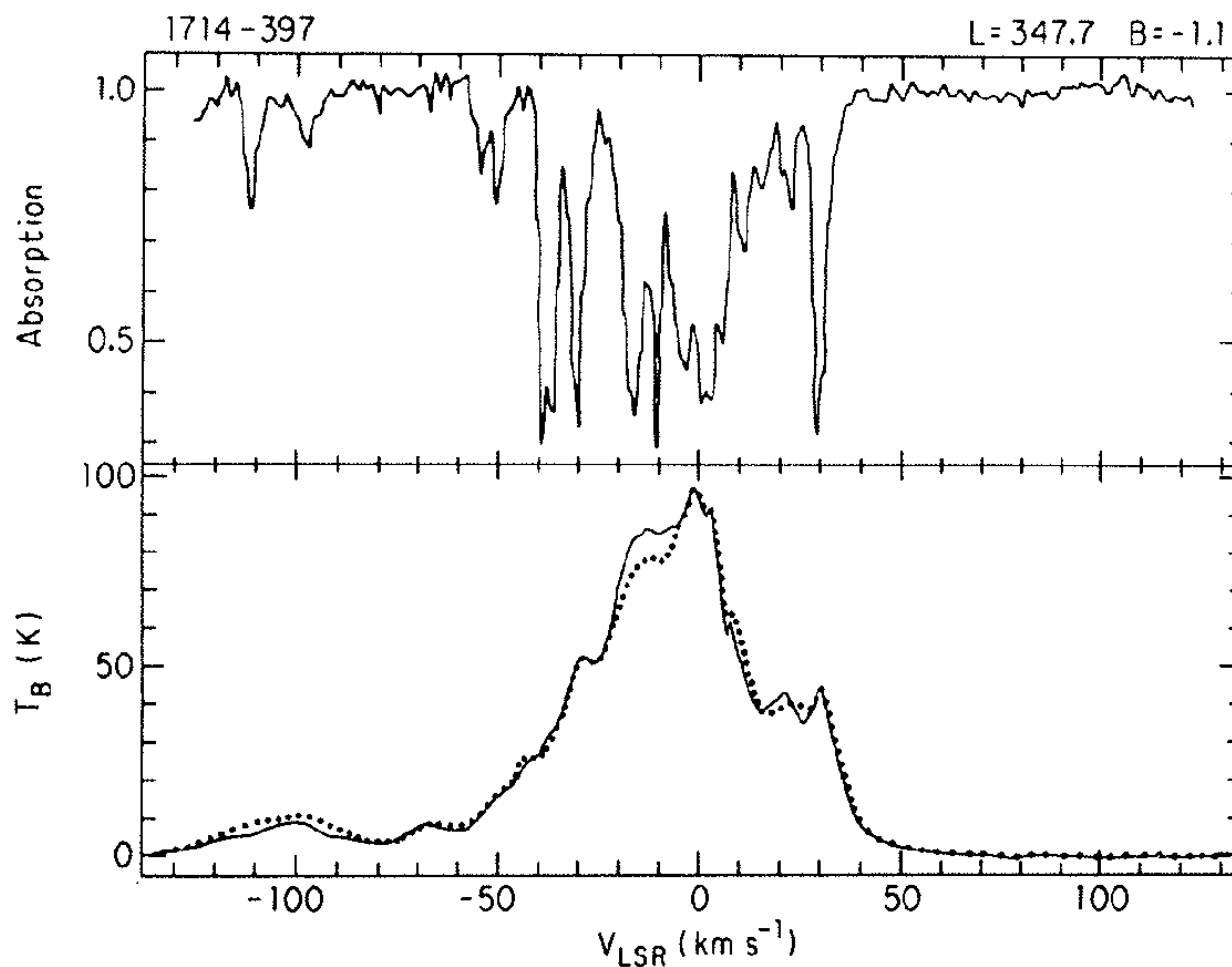
$$N_{\text{H}} \approx \frac{32\pi}{3c^2} \frac{\nu_0}{A_{ul}} \frac{kT_s}{h} \int \tau_{\nu} d\nu$$

In the Rayleigh-Jean limit, $h\nu_0/(kT_s) \ll 1$, if $\tau_{\nu} \ll 1$, then the observed brightness temperature

$$T_b \approx T_s \tau_{\nu} \quad \Rightarrow \quad \tau_{\nu} \approx \frac{T_b}{T_s}$$

$$\Rightarrow N_{\text{H}} \approx \frac{32\pi}{3c^2} \frac{\nu_0}{A_{ul}} \frac{k}{h} \int T_b d\nu \approx 1.82 \times 10^{18} \int \left[\frac{T_b(v)}{K} \right] d\left(\frac{v}{\text{km s}^{-1}} \right) \quad (\text{cm}^{-2})$$

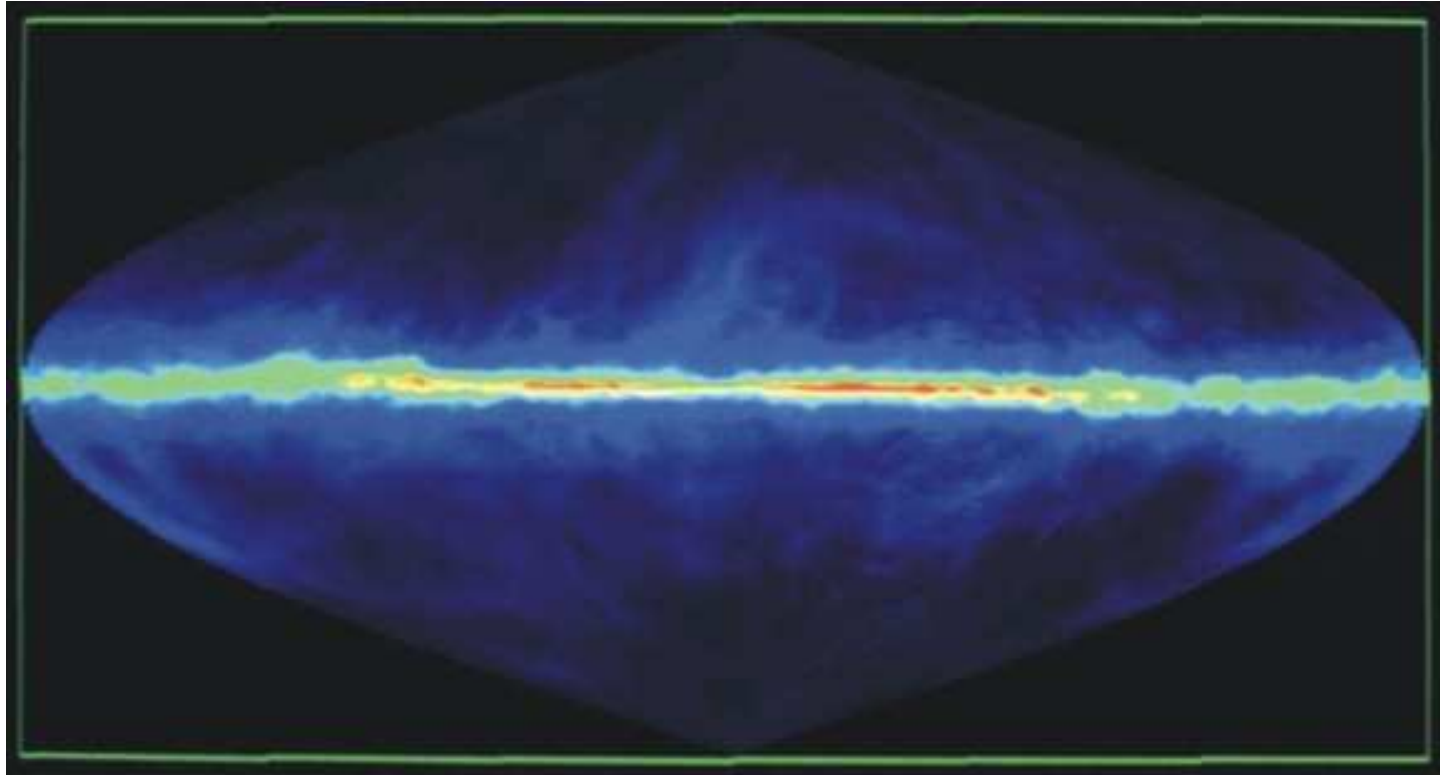
21 cm line toward 1714-397



The HI absorption and emission spectra toward the source 1714-397 (Dickey, J. M. et al. 1983, ApJS, 53, 591).

Galactic Structure in 21 cm

HI clouds and the structure of the Milky Way:

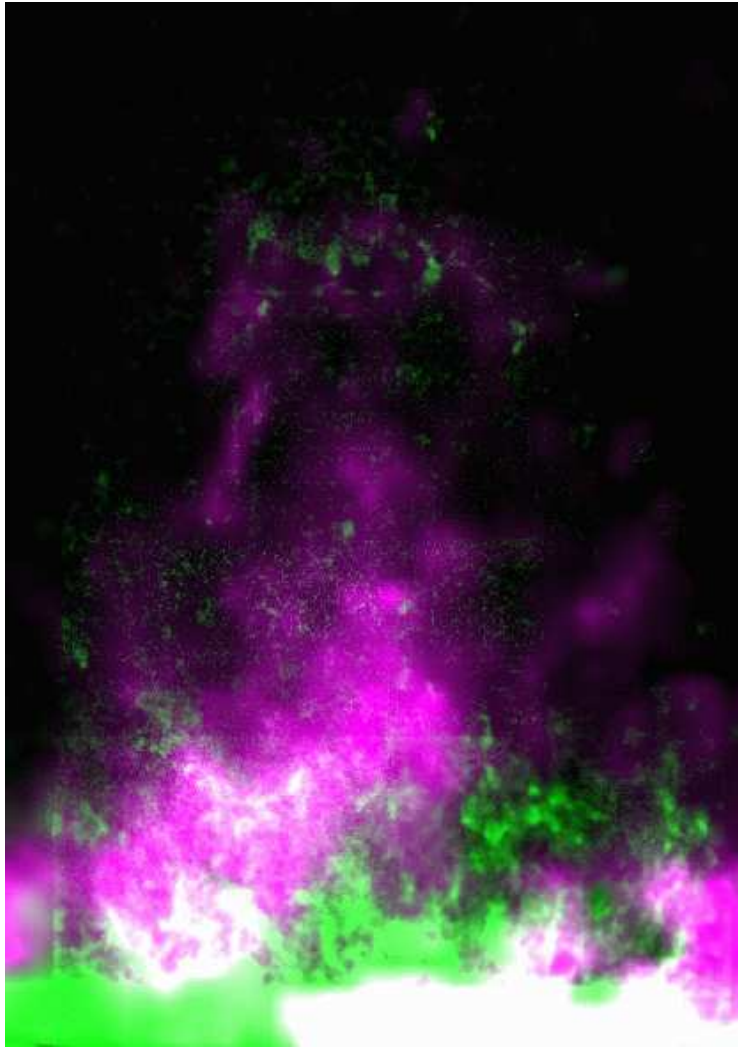


Galactic Structure

The ISM structure of the Milky Way from HI and HII lines:

- temperature is determined by gas heating and cooling: $\frac{d}{dt} \left(\frac{3}{2} n k T_K \right) = \Gamma - \Lambda$
 - Γ : heating (ionization photons from hot stars, high-energy particles [cosmic rays], collisions of gas clouds or streams)
 - Λ : cooling (heavy element spectral emission) e.g., CII at $\lambda = 157.5 \mu\text{m}$
- 4-component model (in pressure balance?) - the swiss cheese picture 
 - CNM (cold neutral medium)
 - narrow 21 cm absorption
 - $T < 50 \text{ K}$, $n > 1 - 10 \text{ cm}^{-3}$.
 - WNM (warm neutral medium)
 - ubiquitous wide 21cm emission
 - $T > 200 \text{ K}$
 - WIM (warm ionized medium)
 - widespread $\text{H}\alpha$ emission
 - $T \sim 10^4 \text{ K}$
 - HIM (hot ionized medium)
 - diffuse soft X-ray emission and O VI absorption
 - $T \sim 10^6 \text{ K}$

Galactic Structure



This image shows a galactic “superbubble” in HI (green) and HII (purple) about 7 kpc distant and 3 kpc in height. Stellar winds and supernovae in young star clusters blow these bubbles. Images of HI away from the galactic plane are easily contaminated by sidelobe responses to the strong and widespread HI emission from the plane itself. The low sidelobe levels of the clear-aperture GBT make such HI images possible (Image courtesy of NRAO/AUI and Yurii Pidopryhora, NRAO & Ohio University)

Extragalactic Studies in HI 21 cm

- Major tool for kinematic studies, such as rotation curve because
 - tracing relative more extended regions because
 - HI - neutral atomic gas
 - low excitation/density requirement
 - spectral (line) feature - kinematic (velocity) probe (with Doppler shifts)

$$v_r \approx c \left(1 - \frac{\nu}{\nu_0} \right) \quad \text{with } \nu = \frac{c}{\lambda}$$

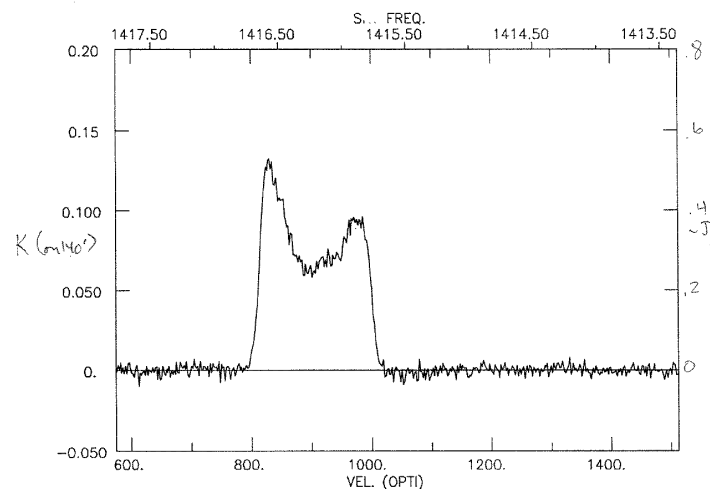
- findings:
 - HI distribution not centered
 - flat rotation curve
 - tidal interaction features

Since 21 cm is such a long wavelength, many galaxies are unresolved by single-dish radio telescopes. For example, the half-power beamwidth of the 100 m GBT is about 9 arcmin at 21 cm. Thus a single pointing is sufficient to obtain a spectral line representing all of the HI in any but the nearest galaxies.

From NRAO Course Astr534

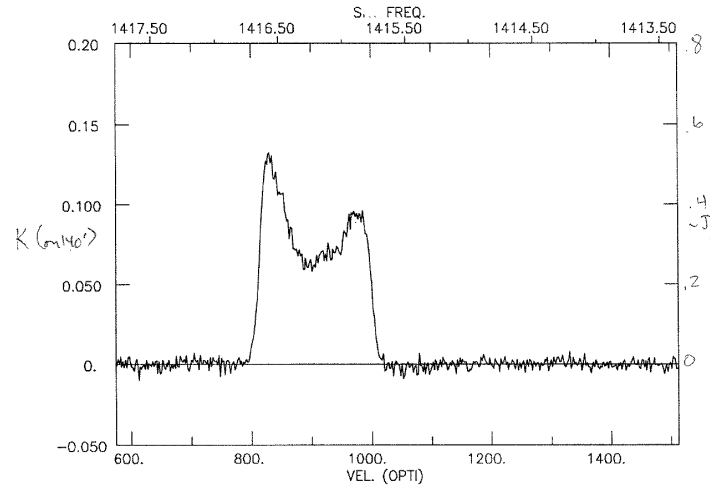
Study **large-scale distribution of galaxies in the universe** because HI is detectable in most spiral galaxies and in some elliptical galaxies.

The radial velocity of a galaxy is the sum of the recession velocity caused by the uniform Hubble expansion of the universe and the "peculiar" velocity of the galaxy. The radial component of the peculiar velocity reflects motions caused by gravitational interactions with nearby galaxies and is typically 200 km s^{-1} in magnitude. The Hubble velocity is proportional to distance from the Earth, and the Hubble constant of proportionality has been measured as $H_0 \approx 72 \text{ km s}^{-1} \text{ Mpc}^{-1}$. **If the radial velocity is significantly larger than the radial component of the peculiar velocity, it can be used to estimate the Hubble distance $D = \frac{v_r}{H_0}$ to a galaxy.**



This integrated HI spectrum of UGC 11707 obtained with the 140-foot telescope (beamwidth 20 arcmin) shows the typical two-horned profile of a spiral galaxy. The radial velocity is $v_r \approx 890 \text{ km/s}$, so $D = 890/72 = 12.4 \text{ Mpc}$.

Extragalactic studies



If the HI emission from a galaxy is **optically thin**, then the integrated line flux is proportional to the mass of HI in the galaxy, independent of the unknown HI temperature. It is a straightforward exercise to derive from previous equation the relation

$$\frac{M_{\text{H}}}{M_{\odot}} \approx 2.36 \times 10^5 \left(\frac{D}{\text{Mpc}} \right)^2 \int \left[\frac{S(v)}{\text{Jy}} \right] \left(\frac{dv}{\text{km s}^{-1}} \right)$$

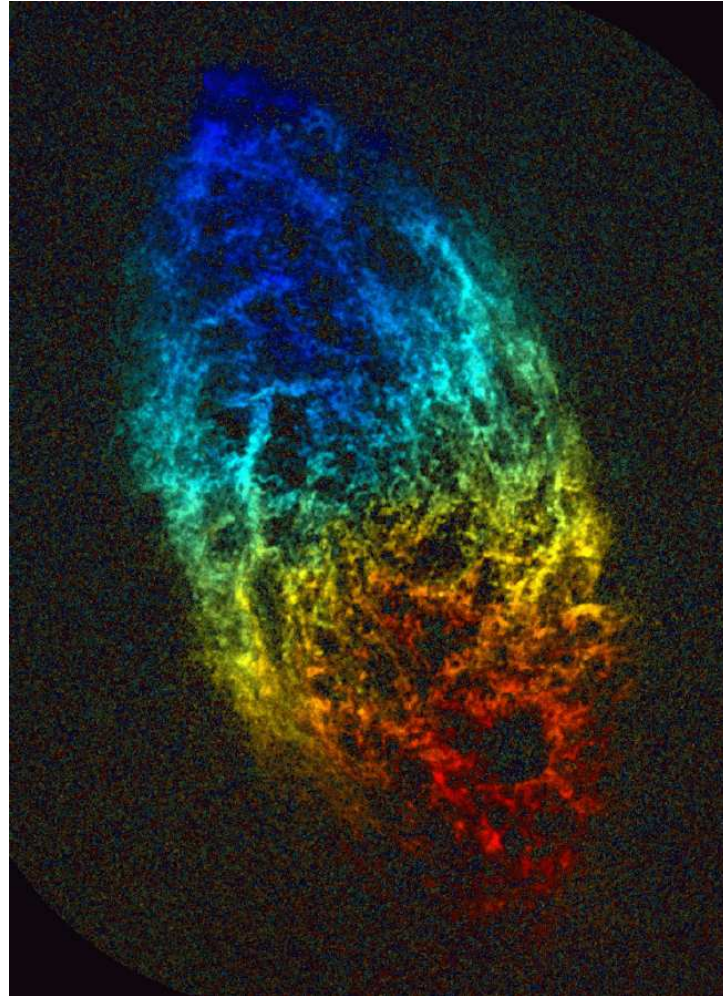
for the **total HI mass** M_{H} of a galaxy. The integral $\int S(v)dv$ over the line is called the **line flux** and is usually expressed in units of Jy km s^{-1} .

Example: What is the HI mass of UGC 11707, assuming $\tau \ll 1$?

From the above spectrum, $\int S(v)dv \approx 0.35 \text{ Jy} \times 200 \text{ km s}^{-1} \approx 70 \text{ Jy km s}^{-1}$.

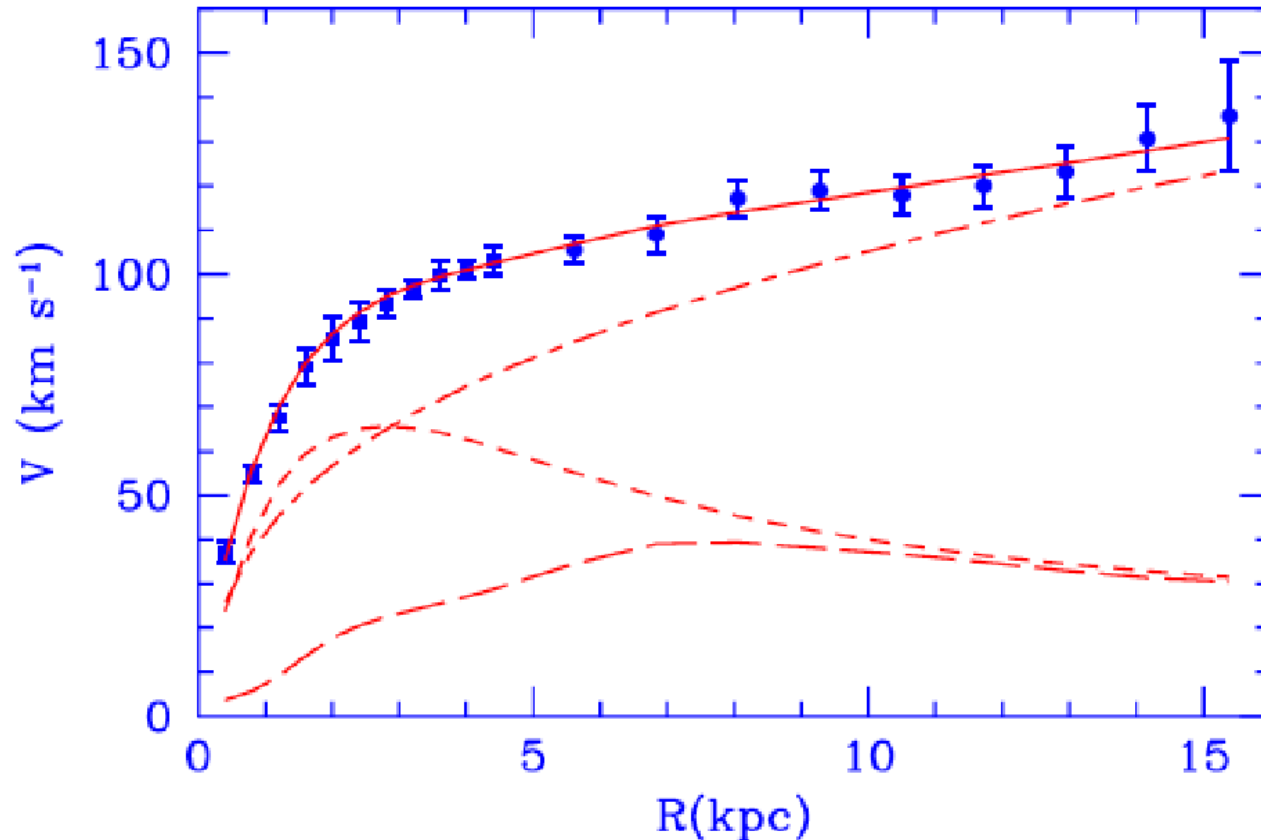
$$\Rightarrow \frac{M_{\text{H}}}{M_{\odot}} \approx 2.36 \times 10^5 \cdot (12.4)^2 \cdot 70 \approx 2.5 \times 10^9$$

The Rotating Disk of M33



(Image courtesy of NRAO/AUI.) The HI radial velocity field of the nearby spiral galaxy M33 is shown here by colors corresponding to Doppler redshifts and blueshifts relative to the center of mass; brightness in this image is proportional to HI column density.

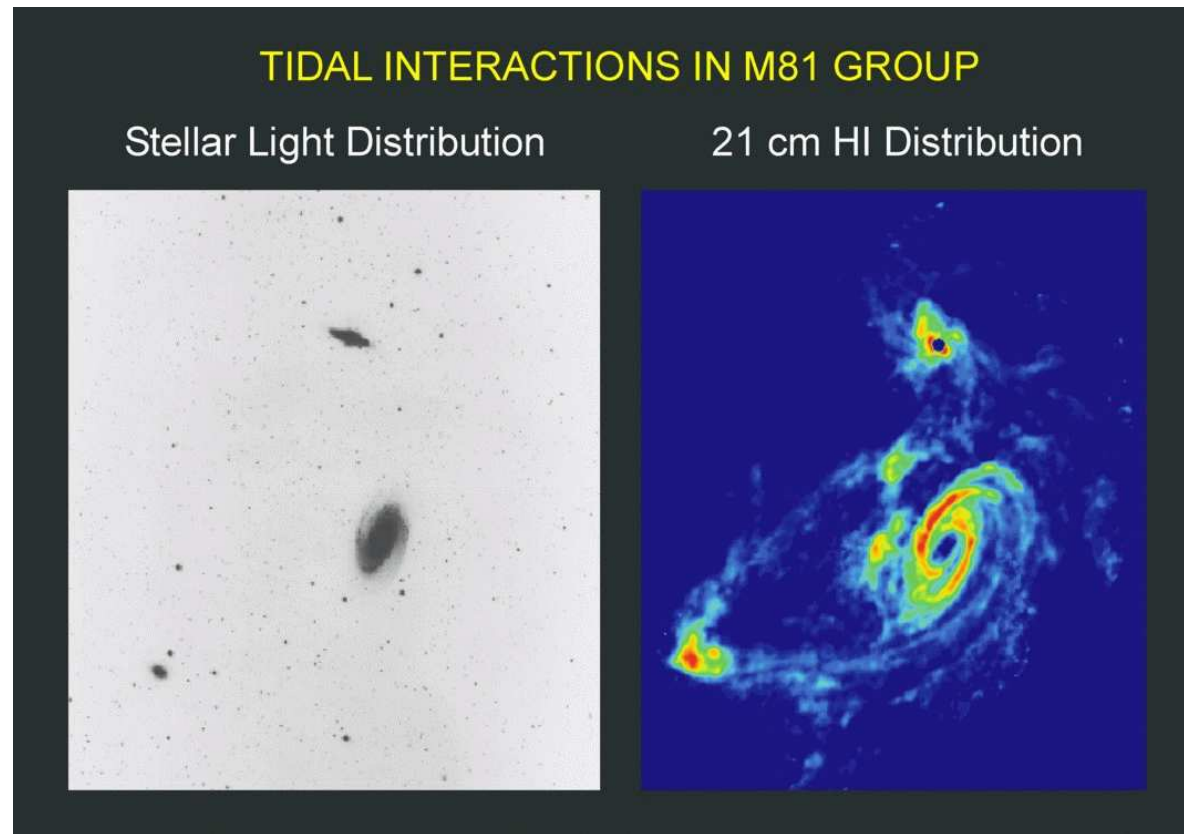
The Rotating Disk of M33



HI Study. M33 rotation curve (points) compared with the best fit model (continuous line). Also shown the halo contribution (dashed-dotted line), the stellar disk (short dashed line) and the gas contribution (long dashed line) [Corbelli & Salucci 2000, MNRAS, 311 441].

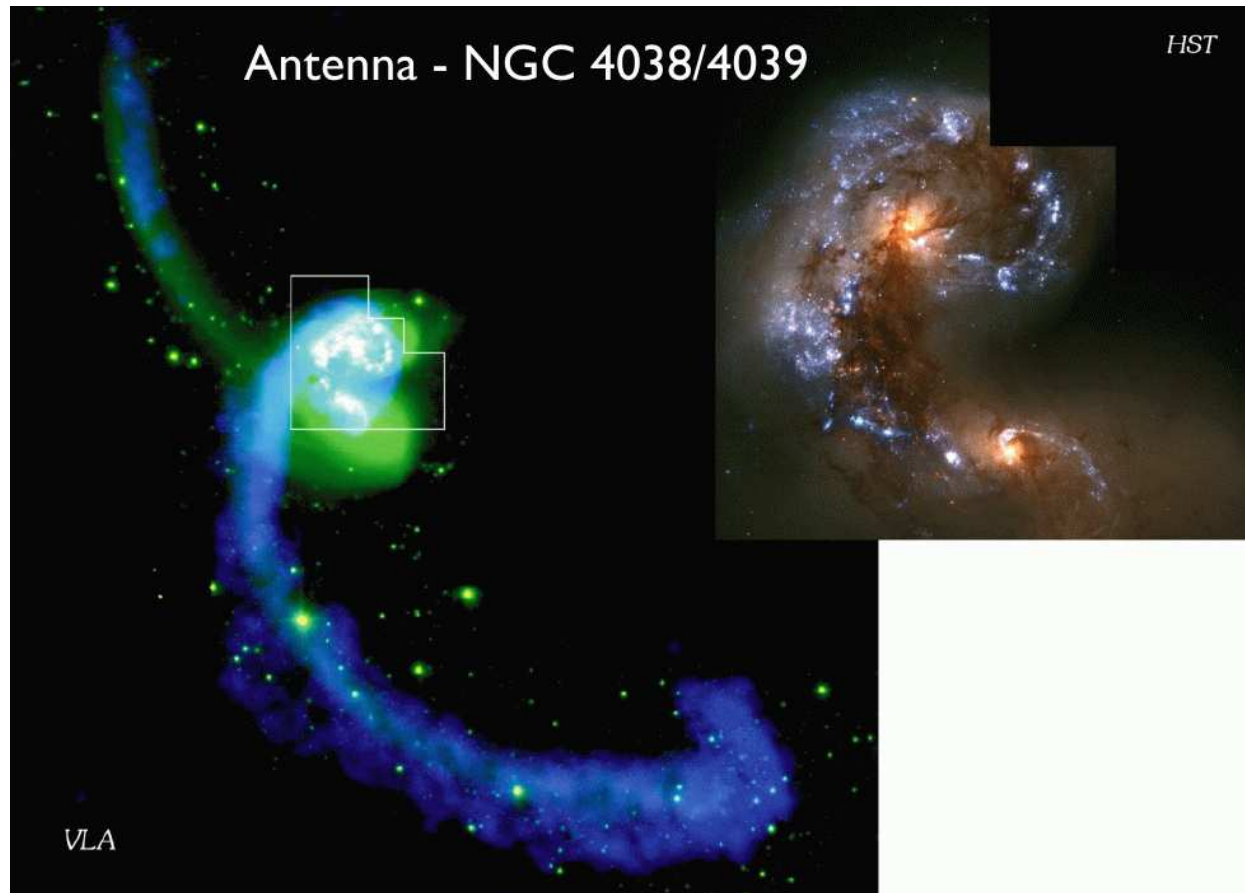
Tidal interactions

Because detectable HI is so extensive, HI is an exceptionally sensitive tracer of tidal interactions between galaxies. Long streamers and tails of HI trace the interaction histories of pairs and groups of galaxies.



(Image courtesy of NRAO/AUI.) The streamers visible only in HI clearly demonstrate that the M81 group is an interacting system of galaxies.

Tidal interactions



(Image courtesy of NRAO/AUI.) Optical (Green) and HI (blue) images of the strongly interacting galaxies NGC 4038 and NGC 4039 (also known as the "antennae"). The velocity distributions of the long HI tidal tails provide strong constraints for computer models of the interaction history.